

Kshitij Bachhav

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EDUCATION

IIIT Nagpur

B.Tech in Electronics and Communication Engineering

Nagpur, Maharashtra

July 2023 – May 2027

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, SQL, C++

Backend: Node.js, Express.js, RESTful APIs

Frontend: React.js, Next.js, Tailwind CSS

Databases: PostgreSQL, MongoDB, Redis, Prisma ORM

Tools: Git, GitHub, Docker, Postman, Render, Vercel

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Authentication (JWT, RBAC, OAuth), Agile/Scrum, Computer Networks

PROJECTS

Pulse: Social Feed Platform | *Express, PostgreSQL, Redis, BullMQ, Socket.io, Next.js* | [Live](#) | [GitHub](#) Jul 2026

- Built a hybrid push/pull fan-out with decay-ranked feeds using Redis sorted sets for write-time ranking and PostgreSQL pulls for high-follower accounts; returned personalized feeds at 74 ms p50 / 92 ms p95, with 8.6× lower p50 latency than per-request ranking, across 5,000 users, 633K follow edges, and 300K posts.
- Designed a BullMQ asynchronous fan-out pipeline sustaining 43,556 follower-feed writes/sec while holding post-creation latency at 50 ms p50, independent of follower count; updated engagement ranking incrementally with a single Redis write per like instead of re-scoring cached feeds.
- Implemented real-time notifications and engagement updates over Socket.io with a Redis adapter for multi-instance delivery, alongside PostgreSQL-native full-text and typo-tolerant search using `tsvector`, GIN, and `pg_trgm`, answering selective queries in 5.7 ms on 300K posts without a separate search service.

Flux-Concurrent Double-Entry Ledger | *Next.js, PostgreSQL, Node.js* | [GitHub](#) Mar 2026

- Engineered a money-transfer service with an append-only, database-enforced ledger where balances are derived, never stored; benchmarked 4 concurrency-control strategies under 200 simultaneous transfers against funds for only 100: naive read-modify-write overdrew the account by Rs. 220 (122 transfers posted), while `FOR UPDATE`, optimistic versioning, and `SERIALIZABLE` each held at exactly 100, with throughput 1.2×–5.9× lower.
- Prevented double-charges under client retries: 100 simultaneous requests sharing one idempotency key produced exactly 1 transfer and 1 debit, with 99 resolved as replays, via a `UNIQUE` key claimed before any ledger write.

Multi-Tenant Project Management SaaS | *React.js, Express.js, PostgreSQL* | [Live](#) | [GitHub](#) Sep 2025 – Jan 2026

- Scoped all 28 REST endpoints by organization ID across 4 tenant layers; cross-tenant reads return 404, not 403.
- Secured the API with JWT, single-use hashed refresh tokens and reuse detection; 3-role RBAC re-checked in the database per request and enforced in React Router guards.
- Delivered Kanban drag-and-drop with optimistic UI, transactional reordering, soft deletes, audit activity logs, and invitation-based onboarding with member lifecycle management.

URL Shortener | *Express.js, Prisma, Redis, Next.js, Docker* | [Live](#) | [GitHub](#) Jun 2025 – Jul 2025

- Generated collision-resistant random short codes over a 62^7 keyspace with retry-on-conflict insertion, avoiding the enumeration and count-leakage risks of sequential ID encoding.
- Served 302 redirects through a Redis cache-aside layer for hot short codes, with a graceful fallback to PostgreSQL when the cache is unavailable; expired links return 410 and are evicted from the cache.

ACHIEVEMENTS

- **Winner, Technex GameJam 2024, IIT BHU** – Ranked 1st among all participating teams in a national-level game development hackathon.
- **DSA:** Solved 200+ problems across LeetCode and GeeksforGeeks, covering arrays, trees, graphs, DP, and greedy algorithms.
- **GitHub:** 25+ public repositories at github.com/zacktiger, with every major project documented with architecture write-ups and reproducible benchmarks.